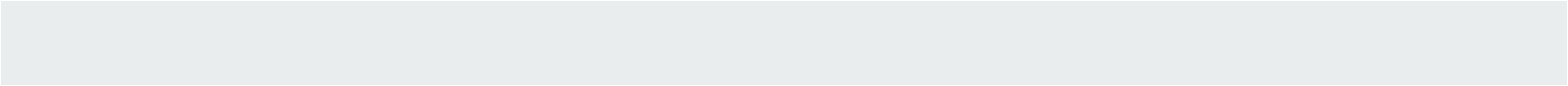




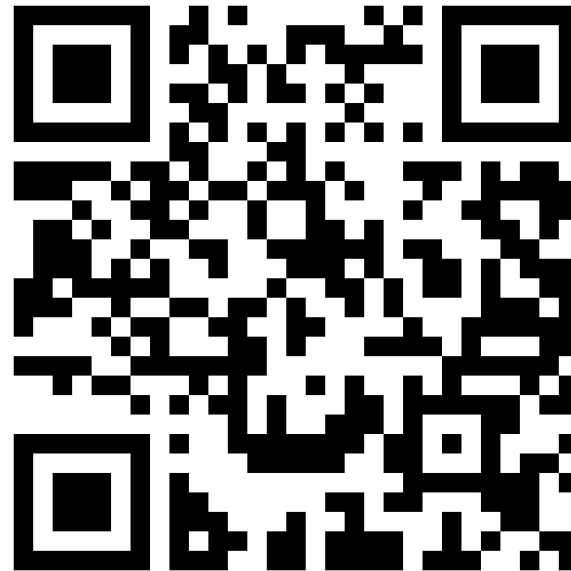
Methodology: Literature Review

BSc Introduction to AI
September 6th, 2024



Before starting...

**Remember to
Join
Slack!**



Next lessons



This week tutoring session:

- Meet your tutor
- Draft ideas for the project
- Submit research problem (Sept 16th)

Next week :

- Methodology lecture (Selin Çakmak) : Group Projects and Team Working
- Guest lecture (Stefan Schlobach): Knowledge Representation

Literature Review

Google Scholar



☒ Articles ☐ Case law

Recommended articles

[Grounded Language Interpretation of Robotic Commands through Structured Learning](#)

A Vanzo, D Croce, E Bastianelli, R Basili, D Nardi - Artificial Intelligence, 2019

[Fast Algorithms for Semantic Association Search and Pattern Mining](#)

G Cheng, D Liu, Y Qu - IEEE Transactions on Knowledge and Data ..., 2019

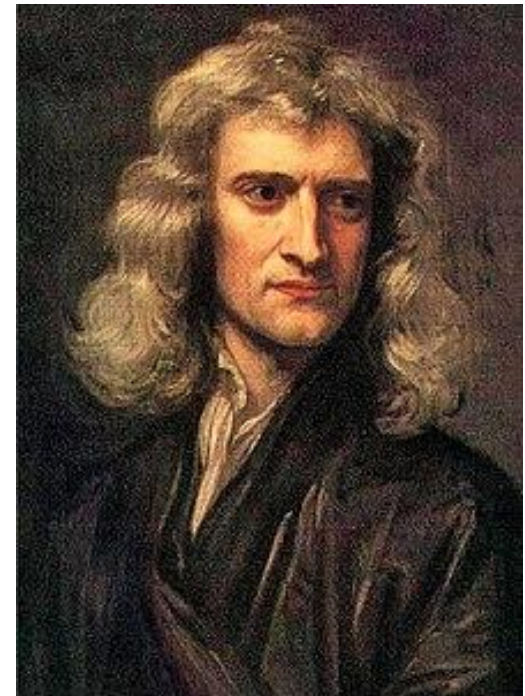
[See all recommendations](#)

Stand on the shoulders of giants

Before Literature Reviews

*“If I have seen further it is by standing **on**
the shoulders of giants”,*

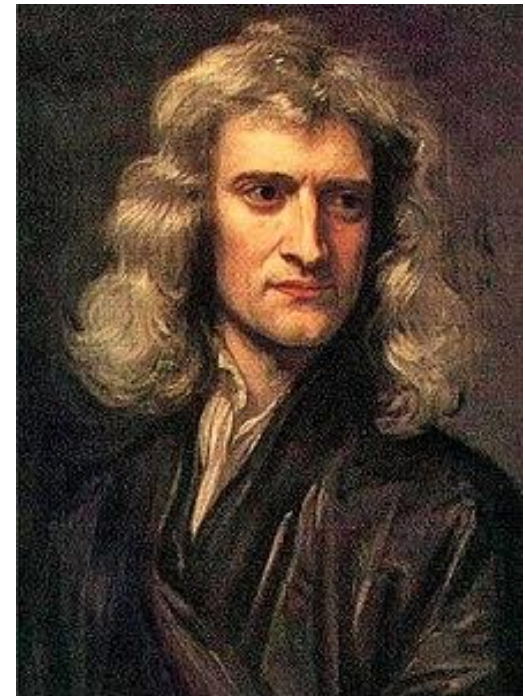
Newton (1675)



Before Literature Reviews

In other words, science can only be
discovered by

building on **previous discoveries**



A generic definition



A literature review is a methodologically **rigorous review of research** results that uses **explicit and reproducible methods**.

Aims

- Fair synthesis of research (without **bias**)
- Rigorously (according to a defined **procedure**)
- Openly (ensuring **visibility** to other researchers)

Do I really need it?



A literature review allows you to find **what is out there**

- Given a research domain, you can find **new research questions**

Do I really need it?



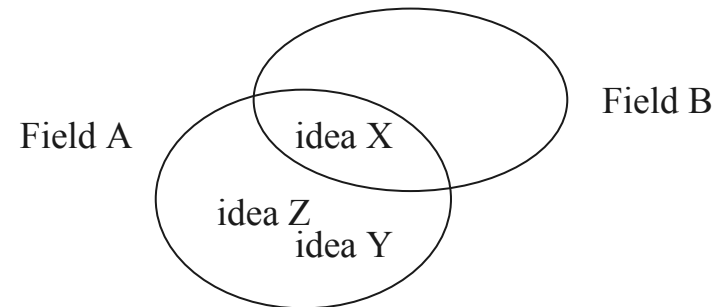
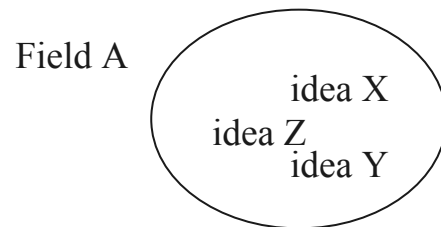
A literature review allows you to find **what is out there**

- Given a research question, you can find an approach to **solve** your issue

Do I really need it?

The literature serves as **foundation** and **support** for new insights

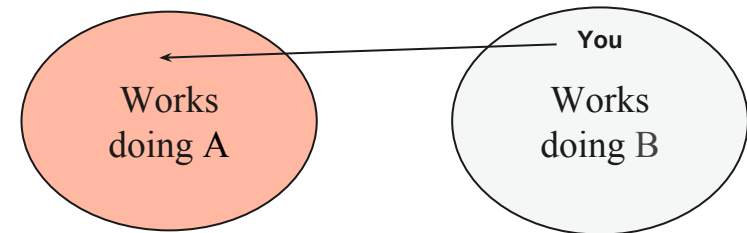
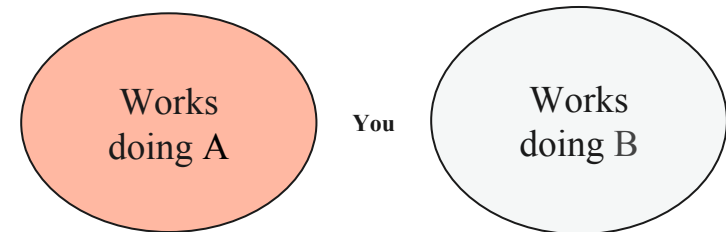
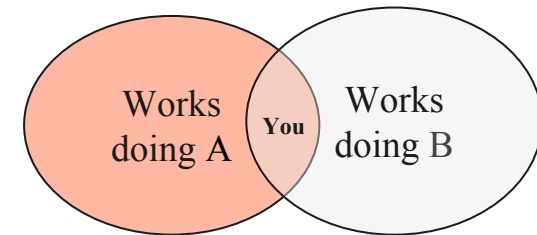
You can pick ideas, criticise, reuse, complain about the others...



Do I really need it?

Show **where** your work lies w.r.t. the others (there are many ways of doing so)

Reveal **commonalities** & **differences** with your own research group



Types of reviews

Systematic reviews

- Use explicit and **rigorous methods** to identify, critically evaluate, and synthesize relevant studies

Narrative reviews

- Summaries of research that **lack explicit descriptions** of systematic methods

Meta-analyses (social/medical science)

- Quantitative review using **statistical** analyses

Types of reviews

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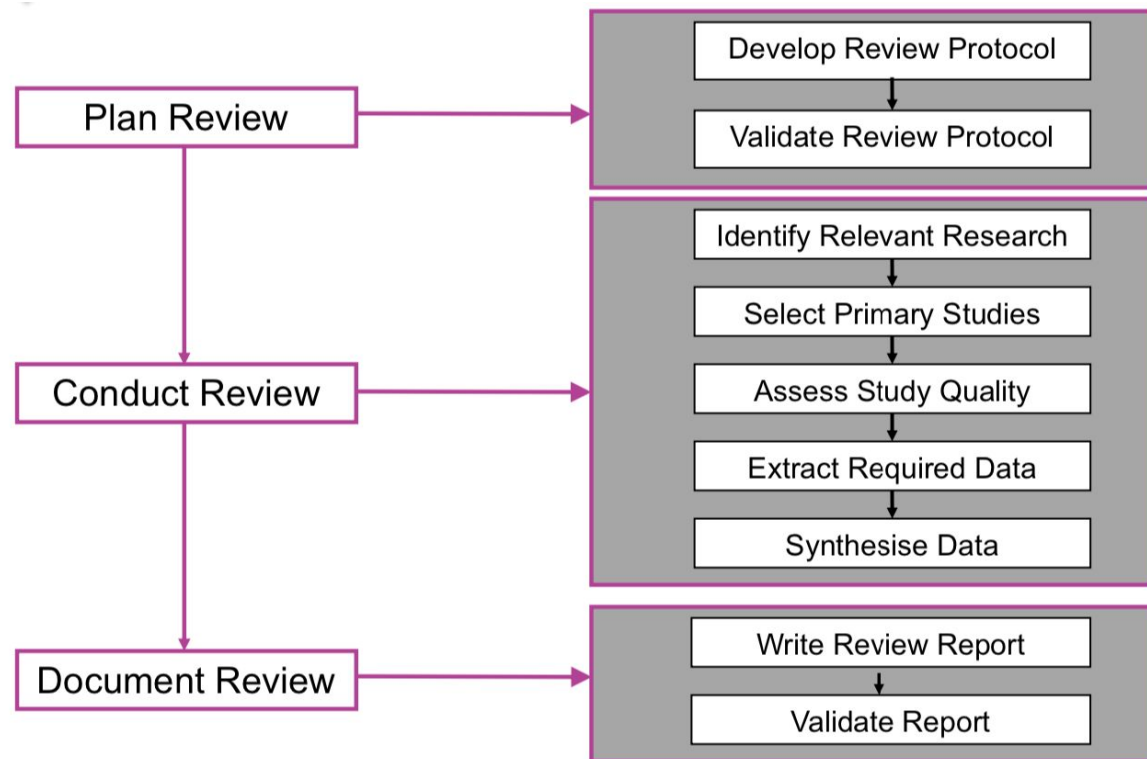
- Quantitative review using **statistical** analyses

How to do a systematic literature review?

1. Plan

1. Conduct

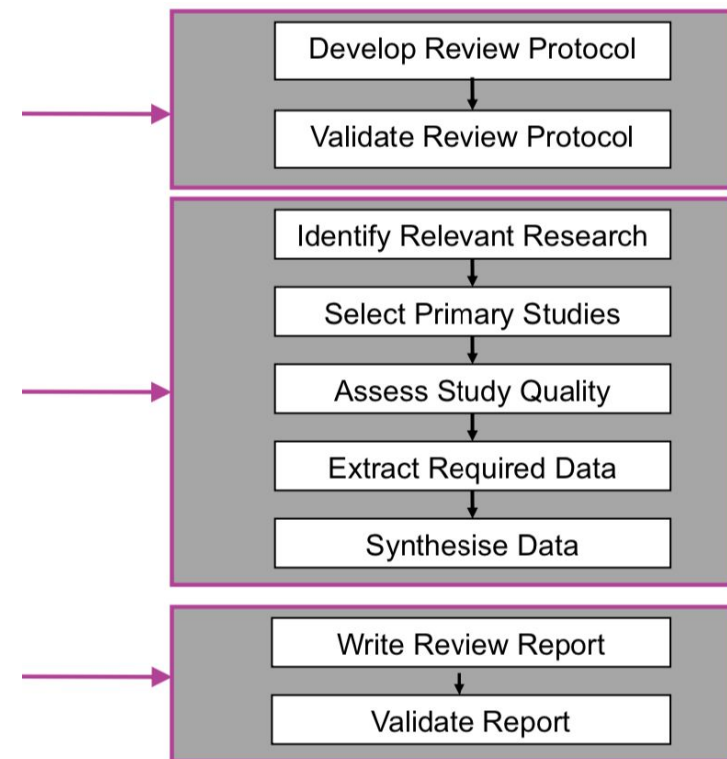
2. Report



How to do a systematic literature review?

In other words, a top-down method

- define a **framework** (characteristics you want to analyse)
- define **search&selection** criteria
- literature **analysis&synthesis** (one-by-one, then organised overview)
- **discuss** what you have drawn and how that helps your work



Where do I start?



Depends on:

- Your research **domain**
Computer Science, Information Science, Artificial Intelligence...
- **Innovativeness** of the topic
if very innovative: perhaps not much published in journals yet
- **Feasibility** (how much time do you have?)
Meta-analyses in bioinformatics can take up to 4 years !

How many papers?



No real answer

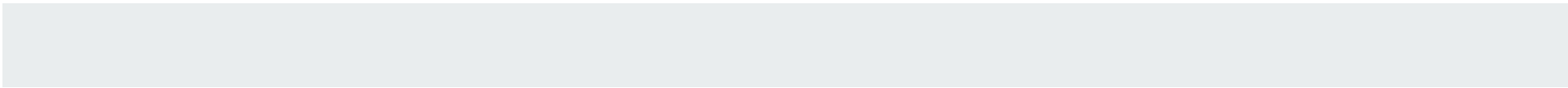
Depends on:

- Your research domain (CS, IS, AI...)
- Innovativeness of your topic
- Type of research paper you are writing
 - research papers/in-use (20-25)
 - systematic literature review papers (50+)
 - posters/demo (3-4)

Protocol for the literature review



- Background & motivation for survey
- Research question(s)
- Search process
- Inclusion and exclusion criteria
- Quality assessment
- Data collection and extraction
- Data analysis
- Report



Setting up things



First, give the context

- Define your background & motivation for the survey
 - e.g. [Tiddi et al, 2019](#), *we have robots + we have smart cities*
 - motivation : *we work on smart cities, but have not integrated robots yet*
- Research question(s) : *what about the integration of robots in smart cities?*

Search : how to



Strategy to find primary studies

- define your **resources**:
select databases, publishers, individual journals and conferences, search engines and search databases
- Define the **search** terms
- Define your procedure for **storing** references
- Decide how **publication bias** will be handled
what to do with the “grey” literature (technical reports, working papers etc) ?
- How **completeness** will be determined?

Search : keywords & queries



Quickly **test** your queries, especially for scope

- Not too broad (unfeasible). Not too narrow (empty net, nothing to learn)

Think of **synonyms**

- “literature review” vs “literature survey”

Think of possible **spellings**

- modelling/modeling, behaviour/behavior

Understand the search engines rules (they have **limitations**)

- to avoid bias

Search : play around with queries...

“adoption of the Internet of Things”

- 778.000 results, too **broad**

What about “adoption **AND** “internet of things”?

- 28.100 results

“user acceptance” **AND** “internet of things” **AND** “survey” **AND** “smart homes”

- 240 results

“user acceptance” **AND** “internet of things” **AND** “survey” **AND** “smart homes” **AND** “nest”

- 43 results, and many irrelevant. too **restrictive**?

Search : ...and document them

List your exact queries in your paper

- Different engines have different syntaxes
- Store your queries (for **repeatability**)
- **Document** your queries in your report

("Business process compliance" OR "Regulatory compliance" OR "Legal compliance")

AND

("Systematic review" OR "Systematic survey" OR "Literature review" OR "Literature survey" OR "State-of-the-art")

AND

("Assess*" OR "Measur*" OR "Monitor*")

Sources



- Online **databases**:

IEEE Xplore, ACM Digital Library, Association for Information Systems

Electronic Library, PubMed

- **Publishers**

SpringerLink, ScienceDirect, Sage, Wiley,

www.ceur-ws.org (open science)...

- **Indexes**

Google Scholar, CiteseerX, Web of Science, VU Lib Search, DBLP

Google Scholar : heads-up



- More results (books, journals, theses, etc)
- Simple to use
- Links to full text items (use UBVU proxy)
- Cited references, related papers
- Less quality control, some non-academic (watch out arxiv.com!)
- Too many results
- Inconsistent bibliographic information

Exercise



- In groups of 5-6
- Search for a term on an engine
- Mark keywords, result numbers, different trials
- What works, what not
- Report to the class your process

Types of literature



- Primary sources = original research
 - (e.g. Research papers)
 - Not filtered through interpretation
 - Not evaluated by second party (e.g. PhD theses)
- Secondary sources = about primary
 - (e.g. other survey papers, text books)
 - selected/re-arranged for a specific purpose or audience
- Tertiary sources = about secondary
 - (e.g. handbooks, encyclopedia)
 - High quality, but (of course) somewhat older

And then what?



Follow the references

- In a paper (outgoing links)
- To a paper (incoming links)
- Related (as in Google Scholar)
- From the same authors

A generic rule



Walk your way up

1. Start with secondary or tertiary sources
2. Use the references to get to (older) primary sources
3. Use “cited-by” search to get to newer primary sources

A generic rule

Use the references to get to (older) primary sources

<https://ceur-ws.org/Vol-2778/paper5.pdf>

...

[18] Musen, M. A. (2015). **The protégé project: a look back and a look forward**. *AI matters*, 1(4), 4-12.

Change Visualization. Katifori et al. [15] and Dudás et al. [4] are two important contributions to ontology visualization. The former covers a range of ontology visualization methods and techniques; it discusses the strengths and weaknesses of each method and addresses the issue of visualizing time-related data. The latter presents the current state of the art in ontology visualization. It states that there is no de-facto standard of visualization which has been accepted by the Semantic Web community, due to the fact that there is no single solution fits all applications.

Visualizations are not new in Protégé [18]. Nonetheless, none of the available plug-ins deal with the direct visualization of changes at editing time. Calculation of impact is also a recent research, and no plugin has addressed it so far. We specifically focus on plugins that visualizes the changes. ChangeAnalysisTab, addition to the Change Management Plugin used in Falconer et al. [6], enables the exploration of changes and annotations using different aspects, such as authors or terms. The Logical Difference Visualizer (LogDiffViz) is noteworthy in terms of its capabilities in comparing ontology versions [8]. Unfortunately, this plug-in does not update visualization based on changes applied during the Protégé session. One can only compare two ontologies and visualize the differences afterwards. Further, OWLDiff [16] does not provide much of a visualization at all. This practical tool serves the comparison and merging of ontologies rather than investigating changes. Lastly, Change View [3] provides a straightforward list of changes but does not visualize them beyond a simple grouping.

A generic rule

Use “cited-by”
search to get to
newer primary
sources

Google Scholar search results for "computation approaches drug repurposing survey". The search bar shows the query and a magnifying glass icon. Below the search bar, it indicates "Articles" and "About 20,700 results (0,14 sec)".

On the left side, there are filters for "Any time" (Since 2023, Since 2022, Since 2019, Custom range...), "Sort by relevance" (Sort by date), "Any type" (Review articles), and checkboxes for "Include patents" and "Include citations" (checked). There is also a "Create alert" button.

The search results list several articles. The first article is titled "[HTML] Review of drug repositioning approaches and resources" by H. Xue, J. Li, H. Xie, Y. Wang, published in the International Journal of Biological Sciences in 2018. It is cited by 512 articles. A red arrow points to the "Cited by 512" link. Other links for this article include "Related articles", "All 9 versions", and "Web of Science: 307".

The second article is titled "In silico methods for drug repurposing and pharmacology" by R. A. Hodos, B. A. Kidd, K. Shameer, published in Systems Biology and Applications in 2016. It is cited by 266 articles. Other links include "Related articles", "All 9 versions", and "Web of Science: 172".

The third article is titled "An overview of computational methods, tools, servers, and databases for drug repurposing" by S. Sarvagalla, S. B. Syed, M. S. Coumar, published in In Silico Drug Design in 2019. It is cited by 21 articles. Other links include "Related articles" and "Web of Science: 172".

The fourth article is titled "A survey of current trends in computational drug repositioning" by J. Li, S. Zheng, B. Chen, A. J. Butte, published in Briefings in Functional Genomics and Proteomics in 2016. It is cited by 482 articles. Other links include "Related articles", "All 15 versions", and "Web of Science: 336".

The fifth article is titled "Computational drug repurposing: current trends" by B. Karaman, W. Sippl, published in Current Medicinal Chemistry in 2019. It is cited by 67 articles. Other links include "Related articles", "All 8 versions", and "Web of Science: 35".

Bibliography Management



Many papers have a unique number

- DOI: Document Object Identifier, e.g. 10.1109/BSN.2012.3
- access online version at the publisher : <http://dx.doi.org/UNIQUE-NUMBER>
- (authors also have a unique ORCID ID: <https://orcid.org>)

Tip : use DOIs to

- Keep track of what you have read
- Keep track of what you want to read

Bibliography Management



Softwares:

- BibDesk (MacOS)
- Zotero (online, MacOS, Linux,.. open source)
- Mendeley (online but proprietary)

etc...

Exchange format for references

- Bibtex (computer science, related to LaTeX mostly)
- APA (social science, use of Word mostly), Chicago, IEEE, etc.

Referencing Papers

S. Keshav. 2007. How to read a paper. SIGCOMM Comput. Commun. Rev.
37, 3 (July 2007). 83–84. <https://doi.org/10.1145/1273445.1273458>

*Author(s). Publication Year. Paper title. Journal/conference
information. Page numbers. Identifier/URL*

NB: Always make sure these info are complete

Other useful tools



Exploring literature:

- ResearchRabbit: <https://www.researchrabbit.ai/>
- ExplainPaper: <https://www.explainpaper.com/upload>
- Connected Papers - to find related work: <https://www.connectedpapers.com/>

Scientific Writing

- LaTeX <https://www.latex-project.org/>
- Overleaf (collaborative/online LaTeX) <https://www.overleaf.com/>

Inclusion and exclusion criteria



Selection criteria allow to **identify studies** that provide **direct evidence** about the research question.

Decide the criteria during the protocol definition
you can **refine them** during the search process.

Inclusion and exclusion criteria

Tiddi et al. 2019

Robot–City Interaction: Mapping the Research Landscape. A Survey of the Interactions Between Robots and Modern Cities

<https://link.springer.com/content/pdf/10.1007/s12369-019-00534-x.pdf>

Finally, the relevance of the works with respect to RCI was assessed by four judges, i.e. a work was considered if at least two of the following conditions applied:

- (1) *Urban robots* the work involves robots operating with some levels of autonomy in a urban space—a social space including a population involved in commercial, administrative, industrial or cultural activities (as per definition of Sect. 2.1);
- (2) *Smart infrastructures* the work focuses on the technological integration of robots in a city ecosystem, and more particularly there is an aspect of data exchange between robots and the city infrastructures;
- (3) *Open-ended interactions* the work studies robots operating in a highly-dynamic context involving interactions of different kinds, and where proactivity in decision-making is necessary to deal with an unforeseeable number of situations.

How to choose a first paper



Look for **surveys**

- papers that conducted a systematic literature review
- get an overview of approaches that can help you determine where to continue to look

Adjust starting keywords:

- computation approaches drug repurposing **survey**

How to choose a survey paper



Select one that captures the most criteria:

- **Cited many times** (survey is already known for a few years)
- Most **recent** (2020 and newer)
- Published in a **journal**, *e.g. ACM Computing Surveys*

Information overload?



How do I read all this?

Save time

- Read the abstract
 - Read the Intro
 - Read the conclusions
 - Still relevant?
 - Read the rest

“Shall I include this work?”



- Does it address your central research problem?
- Does it address a related problem ?
- Did the work identify the problem first?
- Does it use the same approach, for a similar problem?
- Can you get inspiration from it in the future?

“How do I judge quality?”



- Link between theory and experiment?
- The right references?
- Correct data acquisition?
- Correct data analysis?
- Correct argumentation?
- Sufficient focus?
- Strong and weak points

Data Extraction

- Think of **what data** will be extracted from each source
 - questions, contributions, open issues
- Use **multiple reviewers** to address data reliability
- **Store** your papers! (Dropbox, Google drive, Mendeley)
- Make **statistics and trends** from data (time/text analysis etc)

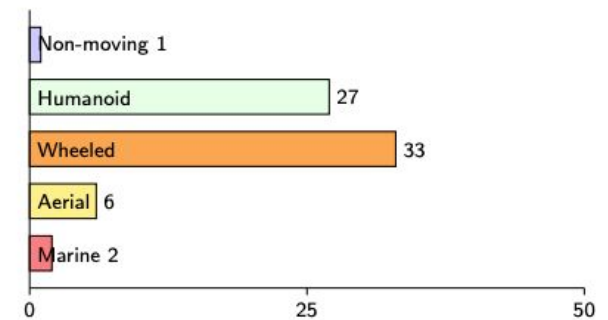
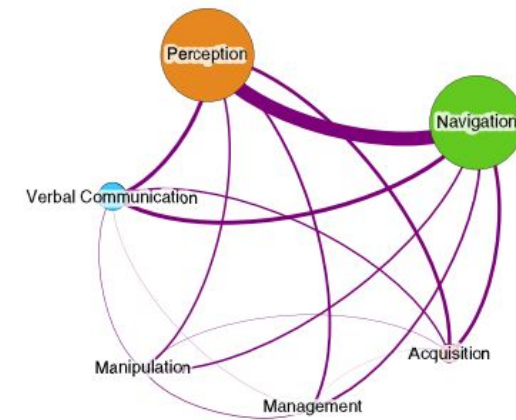
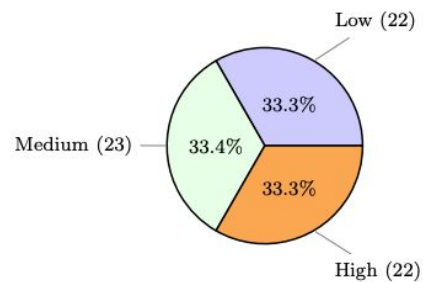
	A	B	C	D	E	F	G	H	I	J	K	L	
1	Instructions >>	choose a paper (if it is red, thour choice in n their category (robots, city, data, interaction))											
2		Name of dimension			R1 - Robot type					R2 - Robot actions			
3	Reference to Papers	Link	Year	Non-moving	Wheeled	hied	not human-f	Aerial	Marine	Navigation	Perception	Management	Manipulation/ ert
7	Cooperative autonomous driving - Int	https://www.researchgate.net/	2005				x			x	x		
8	Dynamic Obstacle Avoidance Appoa	https://www.researchgate.net/	2012				x			x			
9	Human-Friendly Networked Partner R	http://ieeexplore.ieee.org/doc/	2006		x					x	x		
10	Human-Robot Interaction in Public an	https://link.springer.com/chap/	2015		x					x	x		
11	Humanoid Robots Rescuing Humans	http://ieeexplore.ieee.org/stan	2015		x	x				x	x	x	x
12	Intelligent Cooperative Control for Urt	http://www.et.byu.edu/~beard	2013					x		x	x		
13	Intelligent Robotic Sensor Agents for	https://pdfs.semanticscholar.c	2002				x			x	x	x	
14	Leveraging open data for supporting t	https://www.researchgate.net/	2016					x		x	x		
15	MINERVA: A Second Generation Mix	http://robot.csl.cmu.edu/kevin.in	1999		x					x	x		

Data Analysis and Synthesis

Choose your **formats** for summarizing data

Use your criteria for their **categorization** and evaluation

Draw **conclusions** : what is there and what not



Data Analysis and Synthesis



What do you want to find out

- What is known?
- What is unknown?
- What is broadly accepted?
- What is controversial?
- What still needs to be done?

Data Analysis and Synthesis

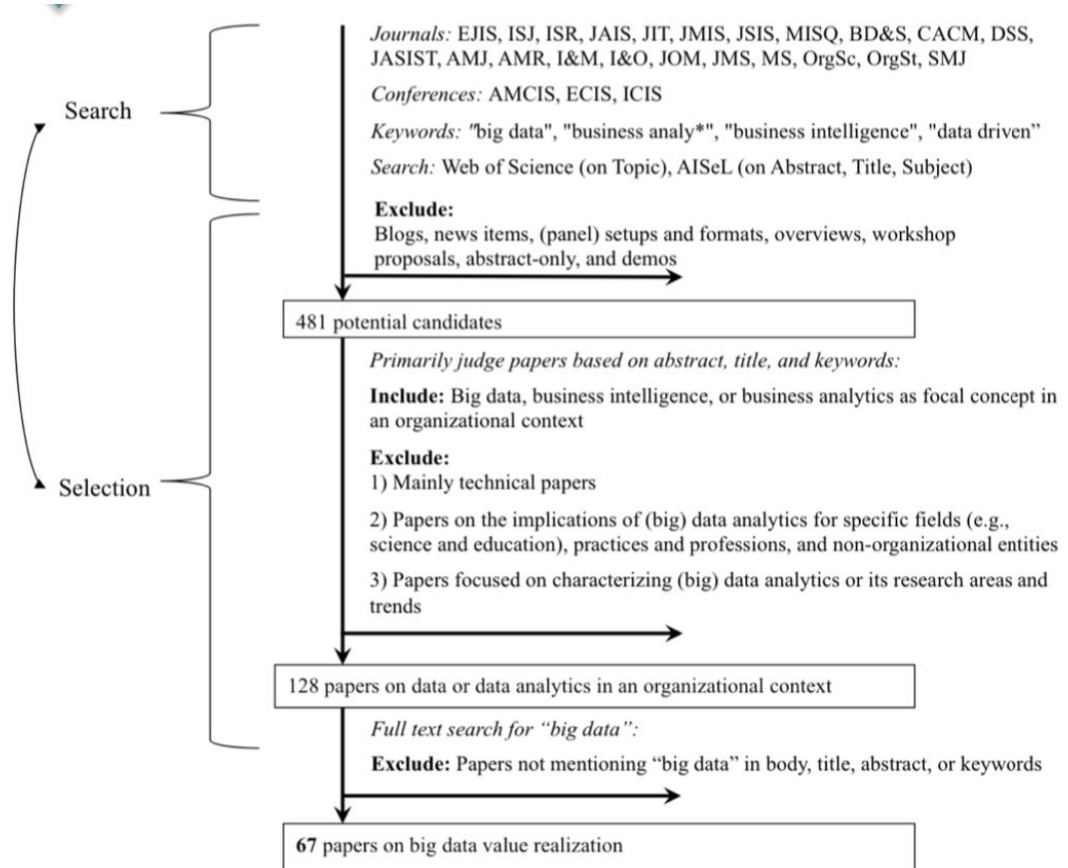


Test your data for **variability**

- Test your queries, inclusion, and exclusion criteria
- What to do get with 4 papers? And with 500 papers?
- Reduce/increase to a **manageable number** of papers (20 to 50)
- Document how many (and which) papers got **added/filtered out**
- Review your decisions in **your team** to avoid bias

Reporting

Include your
method in your
work (graphically,
or text)



Include your results

Table 3 Overview of the literature analysis

[illegible]

Reporting



Organise papers in **specific bodies** (subareas)

Narratively show how the body **holds together**

- in some philosophical or technological way
- or : by publication, by method, by trend, thematically...

Characterise each group :

- Find problems, methods, contributions, results, failures, etc...

Reporting



Discuss on the **limitations** of your literature reviews

Publication bias

- Negative findings are hardly published
- Positive findings are more than once

Selection bias

- Inclusion and exclusion criteria can create bias

Reporting

- Introductory sentence
- Paper 1 description.
- Paper 2 description.
- More paper descriptions...
- How is your work **different** from the above mentioned papers.

A well-known consequence of evolution is semantic drift or semantic impact. SemaDrift [22] is a tool to calculate various semantic drift measures between versions of ontologies. It applies different methods of calculation and distinguishes between an exact, inexact, and hybrid ontology matching approach. In OntoDrift [5], the authors introduce an update to semantic drift measures used in SemaDrift for a better global comparison. In contrast to these tools, we compare the impact of ontology change in the context of embeddings, which can then be used in multiple more specific tasks like link prediction, classification, or recommender systems, instead of the ontologies directly.

There are other studies that directly investigate the effect of ontology evolution on a specific task. Gonçalves et al. [8] define a categorization of changes based on entailment impact. For the classification of a change, they investigate if it influences the set of entailed axioms in the next version. Gross et al. [10] examine how the changes in an ontology impact previously conducted functional analysis. They define an impact measure based on stability of concepts over time and investigate the results of enrichment analysis using real and synthetic data to prove the effectiveness of the measure. Gottron and Gottron [9] also investigate the impact of KG evolution using LOD. They implement twelve different indexing methods and evaluate how the indices are affected by the evolution of the data using three different measures. Osborne and Motta [14] present the pragmatic ontology evolution, in which they analyze the selection of concepts for a new version by evaluating the performance of four different tasks. Our work, in contrast, focuses on the direct comparison of embedding models.

Reporting



Make some **critical assessments**

Common, but very wrong:

Algorithms A, B, C exists, A is good because of X, bad because of Y, B is good because of Y, bad because of X

Where is the **perspective**?

The elephant in the room

Should I use ChatGPT?



The elephant in the room



Should I use ChatGPT?

- It's OK to ask for definitions and rephrasing, but check the output
- Do not generate references, hallucinations are likely (a paper that does not exist)
- And always check

[https://vu.nl/en/student/examinations/generative-ai-your-use-our-expectation](https://vu.nl/en/student/examinations/generative-ai-your-use-our-expectations)

s

Other Do's and don't's



- Refrain from **abusing quotes**
- Paraphrasing is okay, but report content accurately
- **Do not copy** : it is plagiarism!
 - Check **these slides** about Academic Integrity
- Do not include work only because they are colleagues/friends
(**citation bubble**)
- Assume the reader knows nothing about the area (often the case)

Other Do's and don't's



- Cite Wikipedia, blog posts, news websites
 - these are not reliable sources
- website : only if it's an official tool/company website
 - Add the link as a footnote, not a reference
 - e.g. “our approach uses the PAL Robotics¹ Tiago”

1. <https://pal-robotics.com/>

Exercise (if times allows)



- Make a group (4-5 people)
- Find a paper / literature review on a topic you like
- Identify
 - Research question(s)
 - Inclusion/exclusion criteria
 - Analysis type
 - Anything they missed ?

Diversity and Code of Conduct



Please be respectful and courteous to everyone!

- for class-related interactions: talk to the teacher(s)
- For resources and support : talk to study advisors
- Concerns/questions : department diversity office
 - every 3rd Monday of the month, 12-1pm (lunch hours)
 - diversity.cs.beta@vu.nl

Diversity and Code of Conduct



Please be respectful and courteous to everyone!

At VU level:

- STORM Diversity Committee (DiversityCie)
 - diversitycie@storm.vu.nl
 - Student-led committee (promoting diversity)
- [Student well being](#)
- [3D](#) office: Diversity and support related issues at the VU level

Code of Conduct



1. **Honesty** : intellectual and personal honesty in learning, teaching, research, and service
2. **Trust** : mutual trust (encourage and support the free exchange of ideas
3. **Fairness** : support fairness in the interactions of students, faculty and administrators
4. **Respect** : honour, value, and consider diverse opinions and ideas
5. **Responsibility** : take action when you encounter wrongdoing
6. **Courage** : stand up for values in the face of pressure and adversity

Ref: International Centre for Academic Integrity. (2013). [The fundamental values of academic integrity](#).